A Systematic Replication analysis on U.S. Banks Post-Recession

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Abstract

This analysis presents a systematic replication of Nippani and Ling's (2021) study to examine if U.S. banks have recovered from the past recession when more time has passed since the 2008 financial crisis. Using FDIC data from the past 40 years, I compared pre-recession banks to post-recession banks to see if there was still a decline in performance. The results show that there was a reduction in performance post-recession as the ROE and ROA decreased after 2009. I have concluded that there was a reduction in bank performance after the recession, which agrees with Nippani and Ling's (2021) article, and that banks have not yet recovered from the financial crisis.

1 | Introduction

Over the past 40 years, there has been a sharp decline in the number of banks. The U.S. went from having over 14,000 banks in 1984 to just under 4,000 in 2024. Boyd and Gertler (1994) believe the decline started as soon as the 1980's with asset size being the predictor for poor bank performance as banks with larger assets took riskier ventures than the smaller banks did and smaller banks, unable to keep up with scaling economies, ultimately merge with the bigger banks. Similarly, Beltratti and Stulz (2011) highlight that reliance on short-

term funding made many banks vulnerable during economic downturns, which lead to bank failures or prolonged recovery periods.

The purpose of this analysis is to determine if there was a decline in performance among banks over the years, especially in the aftermath of the recent recessions. There have been studies like this before. So, I chose to do a systematic Replication on the Nippani and Ling (2021) article. Focusing on their first analysis, which aimed to see if there was a decline in bank performance after the 2008 recession. This was to see if a longer period of time or the differences in our methodology have an impact on the end results.

Hypothesis

Null Hypothesis (H_0): there is no significant difference regarding performance between banks in the pre-recession era and banks in the post-recession era.

Alternative Hypothesis (H_1): Bank performance has declined post-recession compared to pre-recession years.

2 | Empirical Analysis

I collected the fifth-quarter banking data from the FDIC API database, which includes the variables Assets, Loan Loss Provisions, Net Interest Margin, Non-Interest income, Non-Interest Expense,

ROA and ROE. Also, I found the annual prime rate from JPMorgan Chase and the RGDP from macrotrends for each year. I sorted each year by asset size then separated them by quintiles into the groups Smallest, Small, Medium, Large and Largest. The Smallest group had the lowest amount of assets, the Small group having the next lowest and so on until the Largest group which are the banks with the most assets for that year.

I used a percentile-based classification instead of fixed asset ranges that most other studies use to provide a more accurate reflection of bank size within each year. Static asset thresholds would not account for inflation and growth of the banking sector over 40 years of time. for example, a top-tier bank in 1984 would not appear in a modern high-asset range, despite being one of the largest banks of its time.

I ran multiple regressions with Loan Loss Provisions, Net Interest Margin, Non-Interest income, Non-Interest Expense, ROA and ROE as my dependent variables and prime rate, RGDP and year (as a binary indicator variable) being the independent variables. I made the years prior to 2010 zero and years from 2010 onward equal one while I excluded the years 2020, 2009, 2008, and 2007 as they were recession years and could cause distortion. The dataset spans from 1984 to 2024 using the fifth quarter data, giving a sample period of 40 years.

The regression model used in this analysis is as follows:

$$Y = \beta_0 + \beta_1(\text{Prime Rate}) + \beta_2(\text{RGDP}) + \beta_3(\text{Year})$$

Where Y represents the dependent variables: Loan Loss Provision, Net Interest Margin, Non-Interest Income, Non-Interest Expense, ROA, and ROE. The RGDP and Prime Rate are control variables to account for the economic conditions and interest rate at the time. The Year is the dummy variable and will show when a dependent variable has significance post 2010.

3 | Data

- Loan Loss Provision

Table 1 is the descriptive statistics and regression for Loan Loss Provision. The Smallest group has the lowest Loan Loss Provisions (-45319) and the biggest Loan Loss Provisions (59315).

The constants for the groups Largest and Smallest are significant with the Smallest being negative and the Largest being positive. This suggests that there were external factors affecting both groups in different ways.

The coefficients for RGDP are positive and significant with Smallest, Large and Largest which means that a majority of the banks' loan loss provision moved in the same direction as the RGDP.

Therefore, when the economy is good, the loan loss provisions grow and vice versa when the economy is bad.

The coefficient for the Prime rate for the Largest group is significant which shows that only the biggest bank's Loan Loss Provision was significantly affected by the Prime Rate.

Lastly, the coefficient for the Year for the Large group is significant and positive, so they increased in Loan Loss Provisions post 2010. However, the variable is not statistically significant for the other groups. So, the conclusion we can draw is that Loan Loss Provisions have not really changed after the recession as only a few banks have experienced changes.

- Net Interest Margin

Table 2 presents the descriptive statistics and regression results for the Net Interest Margin. The smallest Net Interest Margin is the Smallest group (-0.70045) and the largest group has the biggest net interest margin (0.54103).

All group constants are positive and significant. This signifies that there are factors, other than the included variables, that affect the net interest margin positively.

The coefficients for the RGDP are negative across all groups, with medium, large and largest being significant. This shows that Net

Interest Margin tends to decline during economic growth, especially for groups with more assets.

The coefficient of Largest group's Prime Rate is positive and significant, so the biggest group's net interest margin had a positive effect with the prime rate coefficient while other banks were not overly affected.

The Year did not have a significant effect on any of the group's Net Interest Margin therefore it was not a variable that was impacted after the recession.

- Non-Interest Income

Table 3 gives the descriptive statistics and regression for the Non-Interest Income. The Largest bank has the lowest non-interest income (-0.979) and the Biggest Non-Interest Income (1.67106).

All constants are positive and significant except for the Largest group which shows there were other factors that positively affected the non-interest income for most banks. The Largest group not being statistically significant implies that either the outside factors had less impact on the Largest group or that their Non-Interest Income is more impacted by the included variables.

The coefficient for RGDP for the Large group is significant and positive. This shows that most of the banks' Non-Interest Income was not affected by changes in the economy.

The Smallest and Small groups have a negative significant prime rate coefficient which means that the smaller banks earn less Non-Interest Income when interest rates rise.

The Smallest group has a negative and significant coefficient for the year so only the Smallest bank groups Non-Interest Income was affected after the recession. The other banks Non-Interest Income were not impacted post-recession

- Non-Interest Expense

Table 4 is the descriptive statistics and regression for the Non-Interest Expense for the Smallest group has the lowest Non-Interest Expense (-64250) and the Small group has the biggest Non-Interest Expense (98590).

The constant is negative for all groups except for Largest group with Smallest and Medium being significant which means that there were other factors that made the Non-Interest Expense negative for most banks.

The coefficients for the RGDP are positive and significant except for the Small group which means that as the economy does well, the Net-Interest Expenses increase and as the economy does worse, the expenses go down.

Only the Smallest group has a significant coefficient for the Prime Rate, so the prime rate only affected the Non-Interest Expenses for smaller banks.

The year is insignificant, so non-interest expense was not affected after the recession.

- ROA

Table 5 is the descriptive statistics and regression for the ROA. The Medium group has the lowest ROA (-0.02208) while the Largest group has smallest ROA (0.68555).

The constant is positive and significant for Smallest, Small and Medium groups which means that there are other factors affecting the ROA, especially with the smaller groups.

The coefficients for the RGDP are positive and significant for all groups, which indicates that during economic growth the ROA increases and vice versa when the economy is bad.

The coefficient for the Prime Rate is insignificant so it does not have a significant effect on ROA.

The coefficient for the Year is negative and significant excluding the Medium group which means that almost all banks' ROA have decreased since the recession.

- ROE

Table 6 is the descriptive statistics and regression for the ROE. The Large group has the lowest ROE (-26.4339) and the Smallest group has the biggest ROE (9.4946).

The constant is positive for all groups but only the Small group is significant so there were not many external factors affecting the ROE for most banks.

The coefficient for the RGDP is positive and significant for all groups showing that during economic growth, the ROE increases for all banks and vice versa for a bad economy.

The coefficient for the Prime Rate is insignificant so ROE and prime rate did not affect each other.

All the Year coefficients are negative and significant except for the Large group, therefore almost all bank's ROE did poorly after the recession.

4 | Conclusion

Looking back to our original hypothesis, we must reject the null hypothesis as there was a significant effect with ROE and ROA declining post-recession. These two variables show a consistent and

statistically significant decrease post-recession, indicating that banks have not fully returned to their pre-recession profitability.

Looking back, while the direction of the results aligns with the results of Nippani and Ling (2021), the severity of the results appears to be less pronounced in this analysis. The differences may be because of our differences in methodology, such as the revised time frame or percentile-based asset groupings used in this replication. Although there were differences in the amount of significant variables that differed between the studies, the central conclusion remains the same: bank performance, particularly in terms of ROA and ROE, has declined in the post-recession era.

Tables

*Note: the first chart shows the number of observations, the mean, the minimum, the median, and the maximum. The second chart shows the regression results with the independent variables in the first column and the results for each group. The period covered is every fifth quarter for every year between 1984-2024. The T-Values are in parentheses. The estimates are above the T-Values and the standard errors are below the T-Values. Significance level is shown as ** $p < 0.05$, *** $p < 0.001$.

Table 1: Loan Loss Provision

Group	n	mean	sd	min	mdn	max
Smallest	41	67388.32	20990	-45319	-26	59315
Small	41	3187.937	5349	-2543.2	-1226.5	29633
Medium	41	1657.721	3057	-1679.1	-141.9	16910
Large	41	641.6797	60.03	-197.8	1.991	107.189
Largest	41	285.1365	50.06	-82.526	-9.777	177.847

Variable	Smallest	Small	Medium	Large	Largest
constant	1.61E+04	4.11E+03	2.35E+03	4.62E+01	3.85E+01
	(-3.505) **	(-0.629)	-(0.455)	(1.076)	(3.936) ***
	-56585.35	-2588.66	-1069.1148	49.68378	1.52E+02
RGDP	9.70E-01	2.47E-01	1.41E-01	2.77E-03	2.31E-03
	(8.465) ***	(1.597)	(0.899)	(16.446) ***	(3.936) ***
	8.211599	0.394945	0.1270831	0.045629	2.04E-02
prime rate	1.76E+05	4.49E+04	2.56E+04	5.03E+02	4.20E+02
	(1.238)	(0.6)	(0.333)	(-1.775)	(-3.950)***
	217965.312	26912.75	8551.5499	-893.727	-1.66E+03
year	1.47E+04	3.74E+03	2.14E+03	4.19E+01	3.50E+01
	(0.771)	(-0.69)	(0.623)	(3.932) ***	(-1.28E+00)
	11304.6932	-2576.77	1329.6591	164.8281	-4.48E+01
R^2	0.8909	0.01649	0.1122	0.9786	3.932

Table 2: Net Interest Margin

Group	n	mean	sd	min	mdn	max
Smallest	41	3.862015	0.3208	-0.70045	0.02696	0.5003
Small	41	4.023123	0.2426	-0.50642	-0.0036	0.4296
Medium	41	4.091329	0.1962	-0.40979	0.02598	0.39895
Large	41	4.13931	0.1806	-0.34018	-0.01277	0.42128
Largest	41	4.152212	0.2	-0.31066	-0.01861	0.54103

Variable	Smallest	Small	Medium	Large	Largest
constant	2.47E-01	1.87E-01	1.51E-01	1.39E-01	1.54E-01
	(17.163)***	(23.445)***	(29.482)***	(31.528)***	(28.152)***
	4.24E+00	4.38E+00	4.45E+00	4.38E+00	4.33E+00
RGDP	1.48E-05	1.12E-05	9.07E-06	8.35E-06	9.25E-06
	(-3.54E-01)	(-1.99E+00)	(-3.315)**	(-3.904)***	(-5.607)***
	-5.25E-06	-2.23E-05	-3.01E-05	-3.26E-05	-5.18E-05
prime rate	2.69E+00	2.03E+00	1.65E+00	1.51E+00	1.68E+00
	(-8.98E-01)	(3.46E-01)	(9.39E-01)	(2.45E+00)	(4.264)***
	-2.42E+00	7.05E-01	1.55E+00	3.71E+00	7.15E+00
year	2.24E-01	1.69E-01	1.37E-01	1.26E-01	1.40E-01
	(-1.62E+00)	(-1.65E+00)	(-1.36E+00)	(-1.30E+00)	(2.48E-01)
	-3.63E-01	-2.80E-01	-1.87E-01	-1.64E-01	3.46E-02
R^2	0.2015	0.5837	0.7171	0.7945	0.8396

Table 3: Non-Interest Income

Group	n	mean	sd	min	mdn	max
Smallest	41	1.249859	0.217	-0.29713	-0.02115	0.65706
Small	41	1.026763	0.1752	-0.21711	-0.04579	0.63978
Medium	41	1.163728	0.1961	-0.37093	-0.01788	0.43616
Large	41	1.05067	0.2515	-0.56106	-0.01236	0.70493
Largest	41	1.954523	0.65	-0.979	-0.05866	1.67106

Variable	Smallest	Small	Medium	Large	Largest
constant	1.67E-01	1.35E-01	1.51E-01	1.93E-01	5.00E-01
	(9.102)***	(9.166)***	(7.664)***	(3.324)**	(1.792)
	1.52E+00	1.24E+00	1.16E+00	6.43E-01	8.96E-01
RGDP	1.00E-05	8.10E-06	9.06E-06	1.16E-05	3.00E-05
	(2.39E+00)	(1.76E+00)	(2.05E+00)	(3.586)**	(2.29E+00)
	2.39E-05	1.43E-05	1.86E-05	4.17E-05	6.88E-05
prime rate	1.82E+00	1.47E+00	1.64E+00	2.11E+00	5.45E+00
	(-2.981)**	(-2.854)**	(-2.59E+00)	(-7.50E-02)	(1.32E+00)
	-5.43E+00	-4.19E+00	-4.25E+00	-1.58E-01	7.18E+00
year	1.52E-01	1.22E-01	1.37E-01	1.76E-01	4.54E-01
	(-3.570)**	(-2.33E+00)	(8.88E-01)	(-1.63E+00)	(-1.60E+00)
	-5.41E-01	-2.85E-01	1.22E-01	-2.86E-01	-7.24E-01
R^2	0.2681	0.1849	0.62	0.322	0.1047

Table 4: Non-Interest Expense

Group	n	mean	sd	min	mdn	max
Smallest	41	196774.8	-64250	-64250	-3773	39980
Small	41	10638.07	17780	-8433	-3940	98590
Medium	41	4361.875	493.7	-1940.18	49.87	635.02
Large	41	2409.492	201.5	-404.48	43.02	481.61
Largest	41	1253.192	164.9	-315.99	-16.61	559.72

Variable	Smallest	Small	Medium	Large	Largest
constant	1.62E+04	1.37E+04	3.80E+02	1.55E+02	1.27E+02
	(-9.633)***	(-0.627)	(-5.034)***	(-0.919)	(2.335)
	-155907.51	-8578.31	-1911.4677	-142.37	296.16903
RGDP	9.72E-01	8.22E-01	2.28E-02	9.31E-03	7.62E-03
	(25.165)***	(1.634)	(19.645)***	(20.701)***	(13.503)***
	24.47238	1.342594	0.4482721	0.192772	0.1029098
prime rate	1.76E+05	1.49E+05	4.14E+03	1.69E+03	1.38E+03
	(2.790)**	(0.617)	(2.245)	(-0.121)	(-2.633)
	492344.017	92061.27	9294.7206	-205.067	-3641.96
year	1.47E+04	1.24E+04	3.45E+02	1.41E+02	1.15E+02
	(1.208)	(-0.797)	(-0.524)	(1.729)	(-2.584)
	17756.6911	-9890.45	-180.5281	243.2751	-297.5365
R^2	0.9857	0.01339	0.9732	0.9815	0.9389

Table 5: ROA

Group	n	mean	sd	min	mdn	max
Smallest	41	0.910380952	0.3704	-14.8959	1.059	9.4946
Small	41	0.922072566	0.2817	-6.1428	0.4365	6.4943
Medium	41	0.946058204	0.2806	-8.1058	0.4407	5.8484
Large	41	0.8550399	0.2989	-26.4339	0.8804	6.7664
Largest	41	0.630572316	0.2592	-11.0221	0.0214	6.6466

Variable	Smallest	Small	Medium	Large	Largest
constant	2.78E-01	2.11E-01	2.10E-01	2.24E-01	1.94E-01
	2.094	3.01**	3.082**	1.013	1.867
	5.81E-01	6.36E-01	6.48E-01	2.27E-01	3.63E-01
RGDP	9.93E-06	7.55E-06	7.52E-06	8.01E-06	6.95E-06
	2.664	2.394	3.411**	4.582***	3.746***
	2.64E-05	1.81E-05	2.57E-05	8.01E-06	2.60E-05
prime rate	2.82E+00	2.15E+00	2.14E+00	2.28E+00	1.98E+00
	0.194	0.64	0.005	1.258	-0.284
	5.49E-01	1.37E+00	9.83E-03	2.28E+00	-5.62E-01
year	2.03E-01	1.55E-01	1.54E-01	1.64E-01	1.42E-01
	(-3.015)**	(-4.152)***	(-3.293)**	(-2.17E+00)	(-3.245)**
	-1.07E+01	-8.09E+00	-7.66E+00	-8.13E+00	-8.778503
R^2	0.185	0.3401	0.2878	0.1745	0.3338

Table 5: ROE

Group	n	mean	sd	min	mdn	max
Smallest	41	9.235079	5.085	-14.8959	1.059	9.4946
Small	41	9.471763	2.791	-6.1428	0.4365	6.4943
Medium	41	8.729187	3.331	-8.1058	0.4407	5.8484
Large	41	7.105505	5.516	-26.4339	0.8804	6.7664
Largest	41	4.450004	4.268	-11.0221	0.0214	6.6466

Variable	Smallest	Small	Medium	Large	Largest
constant	3.91E+00	2.15E+00	2.56E+00	4.12E+00	2.98E+00
	(1.594)	(2.760)**	(2.075)	(0.759)	(1.152)
	6.23E+00	5.93E+00	5.32E+00	3.13E+00	3.4336349
RGDP	2.35E-04	1.29E-04	1.54E-04	2.48E-04	1.79E-04
	(3.231)**	(4.580)***	(4.076)***	(2.999)**	(3.915)***
	7.59E-04	5.91E-04	6.28E-04	7.43E-04	0.000701
prime rate	4.26E+01	2.34E+01	2.79E+01	4.49E+01	3.25E+01
	(-8.49E-01)	(-4.76E-01)	(-8.25E-01)	(-7.54E-01)	(-2.06)
	-3.62E+01	-1.12E+01	-2.31E+01	-3.39E+01	-66.90938
year	3.55E+00	1.95E+00	2.33E+00	3.74E+00	2.71E+00
	(-3.015)**	(-4.152)***	(-3.293)**	(-2.17E+00)	(-3.245)**
	-1.07E+01	-8.09E+00	-7.66E+00	-8.13E+00	-8.778503
R^2	0.185	0.3401	0.2878	0.1745	0.3338

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